GG-S-928b
MAY 2, 1958
SUPERSEDING
Int. Fed. Spec. GG-S-00928a (GSA-FSS)
May 11, 1956, and
Fed. Spec. GG-S-928
February 28, 1946

FEDERAL SPECIFICATION

SYRINGE, METAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers a metal syringe for use in testing injuries and diseases of the ear.

1.2 Classification.

1.2.1 *Sizes.*—Metal ear syringes shall be of the following sizes, as specified (see 6.1):

Size A.—Two-ounce capacity.
Size B.—Four-ounce capacity.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG-I-526 — Instruments, Dental and Surgical; General Specification.
PPP-B-566 — Boxes, Folding Paperboard.
PPP-B-585—Boxes; Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standard:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-4229 — Boxes; Paperboard, Metal-Stayed.
MIL-B-10377—Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL-L-10547 — Liners, Case Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.

(Copies of Specifications and Standards required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 *Metal parts.*—Metal parts shall be class 4, composition N conforming to the applicable requirements of Federal Specification GG–I–526.

3.1.2 *Washers.* — Shall be of a material suitable for the purpose intended.

3.2 Threads.—All threaded parts shall be threaded with American National Fine Thread series threads.

3.3 Marking.—The name or trade-mark of the manufacturer shall be stamped on the syringe.

3.4 Description.—The syringe shall be fabricated of brass, as specified in 3.1.1, except that plunger proper and inside surface of barrel may be nickel plated only, and shall be of the design illustrated in figure 1, and shall consist essentially of the following: The barrel, a cap with finger grips, the plunger assembly, a guard and two tips, (1 bulbous ear tip and 1 Pomeroy ear tip), (see 6.5). The syringe shall be compact and shall have a working capacity of 2 or 4 fluid ounces for sizes A and B, respectively.

3.5 Construction.

3.5.1 *Barrel.*—Shall be cylindrical in shape with a wall thickness of approximately 0.045 inch. The distal end of the barrel shall be closed except for a centered circular opening approximately ¼ inch in diameter which shall be drilled and tapped for the reception of the threaded ends of the syringe tips and the guard. The inside surface of the barrel shall be accurately machined to receive the plunger, and shall be free from pits, grindmarks, and all other defects. The proximal end of the barrel shall be open, slightly reamed on the inner edge to permit ready insertion of the plunger and threaded on the outside surface for the reception of a threaded cap.

3.5.1.1 *Inside dimensions.*

3.5.1.1.1 *Size A.*—The inside dimensions of the barrel shall be approximately 1½ inches in diameter and 3⅝ inches in length.

3.5.1.1.2 *Size B.*—The inside dimensions of the barrel shall be approximately 1⅝ inches in diameter and 4¼ inches in length.

3.5.2 *Plunger assembly.*—Shall consist essentially of the plunger proper, plunger rod, and thumb rest.

3.5.2.1 *Plunger proper.* — The plunger proper shall be of either type I or type II construction as specified below.

3.5.2.1.1 *Type I.* — The type I plunger proper shall be of heavy walled construction, cup shaped, and shall be approximately ¾ inch in length. The sides of the plunger shall accurately fit and move freely throughout the full excursion of the barrel. The lower edge of the plunger shall be slightly beveled to permit ready insertion in the barrel. The base

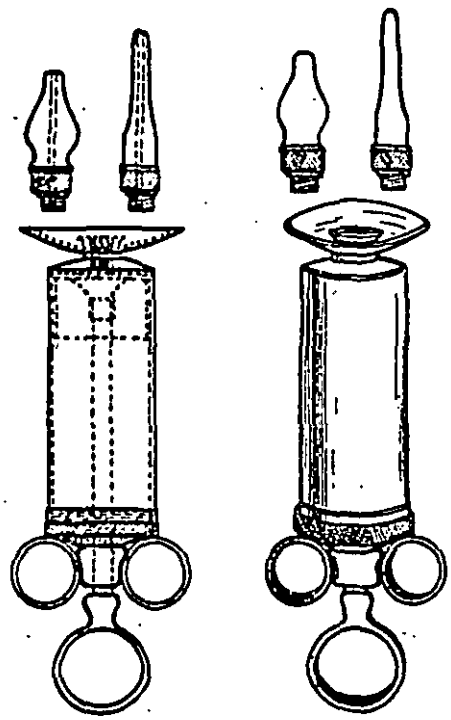


FIGURE 1.—Syringe, ear, metal (Pomeroy), with tips.

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on the inside of the plunger shall be designed and constructed for the secure attachment of the end of the plunger.

3.5.2.1.2 *Type II.*—The type II plunger proper shall consist of a solid, circular cylinder approximately 3/8 inch in depth. The center of the outside surface of this cylinder shall be recessed to properly accommodate an O-ring of adequate size. The sides of the plunger with the O-ring in place shall accurately fit and move freely throughout the excursion of the barrel. The proximal surface of the plunger proper shall be designed and constructed for the secure attachment of the end of the plunger rod. Two extra O-rings shall be supplied with each syringe using this type plunger.

3.5.2.2 *Plunger rod.*

3.5.2.2.1 *Size A.*—Shall be not more than 4½ inches long and ¼ inch in diameter. The rod shall have graduation lines at 0, 1, and 2 ounces and the graduation lines marked to designate 1- and 2-ounce capacity. With the plunger proper fully inserted in the barrel, the 0 graduation line shall be even with the top edge of the guide portion of the cap.

3.5.2.2.2 *Size B.*—Shall be approximately 5¾ inches long and ¼ inch in diameter. The rod shall have graduation lines at 0, 1, 2, 3, and 4 ounces and the graduation lines marked to designate 1-, 2-, 3-, and 4-ounce capacity. With the plunger proper fully inserted in the barrel, the 0 graduation line shall be even with the top edge of the guide portion of the cap.

3.5.2.3 *Thumb rest.*—Shall be a ring approximately 1¼ inches inside diameter 3/8 inch wide and 1/8 inch thick. The thumb rest shall be securely attached to the upper end of the plunger end.

3.5.3 *Cap.*—Shall be of conventional design, threaded on the inside surface for secure and removable attachment on the proximal end of the barrel. A centrally located opening shall be provided, which shall serve as a guide for the plunger in its excursion in the barrel. Two metal finger rings approximately 7/8 inch in inside diameter shall be securely and permanently attached to the cap and/or to an integrally formed tubular extension of the cap.

3.5.4 *Guard.*—Shall be a concave disk, approximately 2½ inches in diameter and of not less than 0.028 inch thickness, provided with a tubular adapter securely and permanently attached in the center, threaded on one end for attachment to the distal end of the barrel with the concave surface toward the tip, and the other end threaded to receive the threaded end of the tips. A washer shall be provided to make a leakproof connection when guard is attached to the barrel of the syringe.

3.5.5 *Syringe tip.* — The tip shall be threaded on one end to fit either the threaded portion of the guard adapter or the barrel. A washer shall be provided to make a leakproof connection. Just below the threaded end, the tip shall be corrugated to facilitate attachment and removal. The lumen of the tip shall be approximately 3/32 inch in diameter.

3.5.5.1 *Bulbous ear tip.*—The bulbous ear tip shall be approximately 1¾ inches over-all length. The tip proper shall be nipple shaped, approximately 1 inch in over-all length. The outer portion of the tip proper shall be approximately ¼ inch long and approximately 9/10 inch in diameter and the inner portion of the tip proper shall be flared outward uniformly to approximately 5/8 inch maximum diameter.

3.5.5.2 *Pomeroy ear tip.*—The Pomeroy ear tip shall be approximately 2 1/10 inches over-all length, of the plain nozzle type. The tapered portion of the tip shall be approximately 1⅜ inches in length and taper from approximately 3/8 inch at the inner end to approximately 1/8 inch at the outer end.

3.6 **Corrosion resistance.**—Syringes shall show no corrosion or continuous bleeding

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from pits when tested as outlined in paragraph 4.3.1. Slight surface corrosion which can be easily washed off shall not be considered cause for rejection.

3.7 Fluid tightness.

3.7.1 *Barrel and plunger.*—The barrel and plunger shall be tested for fluid tightness as specified in 4.3.3.1. During the first and second tests, when pressure is applied as specified in 4.3.3.1, there shall not be enough leakage to accumulate more than 3 cubic centimeters of water.

3.7.2 *Tips.*—The tips shall be tested for fluid tightness as specified in 4.3.3.2. During the first and second tests, when pressure is applied as specified in 4.3.3.2, there shall not be enough water to pass the tip at its threaded connection to produce a falling drop.

3.8 Workmanship. — Shall be first class throughout. The syringes shall be free from all defects which will detract from their appearance or impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *For visual examination.*—Sampling procedures for visual examination shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105. The acceptable quality level for major defects shall be 1.0 percent defective and for minor defects shall be 2.5 percent defective.

4.1.2 *For salt-spray and fluid tightness tests.*—Sample size for tests shall be as indicated for inspection level L4 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero (0) for all sample sizes, and a minimum of five samples shall be taken.

4.2 Inspection.—The inspector shall ascertain that the metal syringes have been inspected and accepted in accordance with specification requirements.

4.3 Tests.

4.3.1 *Salt-spray test.*—Plated parts shall be tested in accordance with applicable section of Federal Specification GG–I–526.

4.3.2 *Chemical composition.* — From the syringes selected for test, millings or drillings for chemical analysis shall be taken to determine conformity of the material with the requirements of this specification. The millings and drillings shall weigh not less than 4 ounces and shall be clean, free from oil, dirt, grit, or other foreign matter. Spectrographic analysis may be used in lieu of the above method.

4.3.3. *Fluid tightness test.*

4.3.3.1 *Barrel and plunger.*—The syringes selected for test shall be thoroughly cleaned with soap and water, rinsed thoroughly and dried. After cleaning, the syringes shall be attached to a testing machine having a water-pressure gage. A pressure of 10 pounds per-square-inch gage shall be applied. The water used as a testing fluid shall be let through the tip of the syringe. The plunger shall be drawn to about the 1-ounce graduation mark where it shall be allowed to remain for 10 minutes. After the first test, any water that may have passed the plunger shall be measured. The plunger shall then be drawn to about the 2-ounce graduation mark, where it shall be allowed to remain a similar length of time, and any water that may have passed the plunger shall be measured. The syringe shall be tested in a horizontal position and shall meet the requirements specified in 3.7.

4.3.3.2 *Tips.*—The test procedure shall be the same as indicated in 4.3.3.1 except that the pressure shall be applied for 10 seconds, and any water that may have formed at the threaded connection of the tip after the first test, shall be wiped away.

4.4 Supplier responsibilities for inspection. —The supplier shall be required to perform the examinations and tests set forth in this specification to substantiate conformance of

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supplies to specification requirements. The examination and tests shall be at the expense of the supplier. Suppliers shall have either adequate test facilities for executing specific prescribed tests, or shall make arrangements for the utilization of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of two years after the delivery of the supplies to which the same relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

5. PREPARATION FOR DELIVERY

(Levels of preservation, packaging, and packing protection used are defined in Fed. Std. No. 102.)

5.1 Packaging.

5.1.1 *Unit of issue.*—One syringe, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.—*Packaging quantities*

Federal Stock No.	Unit package	Intermediate package	Shipping container
6515–879–5850	1 unit	Not required	48 units

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specifications PPP–B–566, PPP–B–676 or Military Specification MIL–B–4229. Carton shall be designed with suitable interior fittings, or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load

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and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Unit package.*—Each unit package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. — Purchasers should specify size required and should exercise any desired options offered herein. (See 1.2.1, 5.1, and 5.2).

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples shall be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured to meet Federal Government requirements.

6.4 **Transportation description.**—Transportation description applicable to this item is:

Surgical instruments, not otherwise indexed by name.

Carload minimum weight 10,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor volume minimum weight 10,000 pounds.

6.5 **For Military Medical Supply Agency procurement.**—Pomeroy ear tip specified in 3.4 is not required.

6.6 This specification covers the following item appearing in the Armed Services Medical Stock List:

6515–379–5850 SYRINGE, METAL, EAR, 2 oz.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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